Parallel Mechanism-Aware Multi-Objective Optimization for Post-Disaster Road Recovery and Relief Logistics

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***Abstract*—Post-disaster road recovery and relief logistics are tightly coupled: repair scheduling determines when the network becomes traversable, while delivery decisions determine how much relief reaches demand nodes and when. Recent models enrich this coupling with complex physical mechanisms—progressive recovery (PR), heterogeneous vehicle accessibility thresholds (HT), and edge-period throughput capacity (EC)—that improve fidelity but make every candidate solution expensive to evaluate through multi-period network simulation. This paper studies not only how to optimize such a realistic recovery–logistics system, but also *when* its complex mechanisms actually matter and *how* their expensive evaluations can be efficiently parallelized. We formalize a three-layer mechanism-aware diagnosis—structural/dynamic exposure, dispatch participation, and objective effect—that separates mechanisms that merely exist from those that change dispatch behavior and, ultimately, optimization objectives. We then present a master–worker parallel multi-objective framework in which fitness evaluation of the full multi-period simulation is distributed across workers while selection, crossover, mutation, random-number generation, archiving, and budget accounting stay centralized, preserving exact reproducibility of the optimization protocol. Diagnostics on the Wenchuan WEN38 network and multi-scale synthetic networks (S025, S050, M100) locate the applicability conditions of PR, HT, and EC. Full-budget NSGA-II runs then show that parallel population evaluation changes nothing but timing: across 136 runs spanning 25 to 100 nodes, every worker count reproduced its reference front exactly—on WEN38, all 40 runs bit-identical to the in-process serial path—while cutting wall-clock time by up to $4.9\times$. Parallel efficiency is governed by evaluation granularity rather than network size, which makes mechanism-aware fidelity reduction and parallel evaluation substitutes rather than complements.**

***Index Terms*—disaster relief logistics; road network recovery; multi-objective optimization; NSGA-II; parallel evolutionary computation; mechanism applicability; progressive recovery.**

I. Introduction

In the aftermath of a major earthquake, two operations proceed in lockstep: crews repair damaged roads, and fleets deliver relief supplies over the recovering network. The two are inseparable—a repair schedule is only as good as the deliveries it enables, and a delivery plan is only feasible on the network that repairs make available. Joint optimization of *road recovery* and *relief logistics* over a multi-period horizon is therefore the natural formulation, following the line of Li and Teo [1].

Realism comes at a price. Three physical mechanisms substantially improve model fidelity. *Progressive recovery* (PR) recognizes that a damaged road is not simply "broken" or "fixed": it passes through blocked, temporary, single-lane, and basic stages, each admitting different traffic. *Heterogeneous vehicle accessibility* (HT) recognizes that a heavy truck and a light van do not share the same passability threshold on a partially recovered road. *Edge-period capacity* (EC) recognizes that each road segment can only carry a bounded throughput per period. Each mechanism is physically plausible, and each makes the evaluation of a single candidate solution more expensive: decoding a solution requires a full multi-period simulation of repair progress, vehicle-specific passability graphs, capacity-constrained dispatch, and fair allocation.

This raises two gaps that existing work rarely addresses together. *First*, it is unclear *when* these complex mechanisms genuinely enter the decision. A mechanism can be present in the model yet never bind: if no delivery ever uses a partially recovered road, PR changes nothing; if fleet capacity is ample, EC is vacuous. Optimizing with a full-fidelity model in a regime where its mechanisms are dormant wastes computation and obscures which physics actually drives the trade-offs. *Second*, even where the mechanisms matter, the sheer cost of full-fidelity multi-period evaluation—repeated for every candidate in every generation of an evolutionary optimizer—dominates runtime and limits the affordable search budget, especially as networks grow.

The two gaps are two sides of one coin. Mechanism-aware analysis asks whether the expensive physics is worth simulating at all in a given regime; parallel evaluation asks how to afford the simulation where the answer is yes. Together they reframe the problem: we study not only how to optimize a realistic post-disaster recovery–logistics system, but also

when its complex physical mechanisms matter and *how* their expensive evaluations can be efficiently parallelized.

This paper proposes a *mechanism-aware parallel* framework that addresses both gaps. Our contributions are fourfold:

1) **A mechanism-aware diagnosis methodology.** We separate three evidence layers—*structural/dynamic exposure* (a mechanism could matter), *dispatch participation* (it changes actual dispatches), and *objective effect* (it changes the optimization objectives)—and establish their logical relations, including where the chain breaks. This turns "does the mechanism matter?" from parameter sweeping into verifiable conditions.
2) **Applicability conditions for PR, HT, and EC.** A diagnostic study spanning 20 fixed decisions on the Wenchuan WEN38 network, 1,120 resource-grid cells across three synthetic scales, and paired relaxation probes locates non-binding, transition, and binding zones for each mechanism—e.g. EC never binds below an edge-utilization of 0.85, with a sharper rerouting-based criterion, and PR exposure is jointly governed by repair time, crew count, and period length.
3) **A master–worker parallel multi-objective optimization framework.** Population fitness evaluation—the expensive multi-period simulation—is distributed across parallel workers, while selection, crossover, mutation, RNG, archiving, and budget accounting remain centralized. The design guarantees that the optimization protocol is unchanged: identical seeds yield identical Pareto fronts for any worker count.
4) **Empirical validation.** We evaluate along four research questions covering mechanism effects (RQ1), applicability conditions (RQ2), runtime reduction without protocol change (RQ3), and what governs parallel efficiency (RQ4). RQ3 and RQ4 are measured with full-budget NSGA-II runs over four instance scales, with frozen seeds—and report exact protocol invariance together with measured speedups; RQ1 and RQ2 characterize the mechanisms through paired decision-level toggles, resource grids and relaxation probes, and we report their inferential scope exactly rather than overstate it.

The remainder of the paper is organized as follows. Section II reviews related work. Section III formulates the v2 multi-period recovery–logistics model with PR/HT/EC and three objectives. Section IV presents the mechanism-aware parallel optimization framework. Section V reports the experimental evaluation, and Section VI concludes.

II. Related Work

a) Post-disaster road recovery and relief logistics.: The coupling of network repair scheduling with relief distribution has been studied as multi-period optimization since Li and Teo [1], who jointly schedule repair work and relief logistics over a damaged network; Maya Duque et al. [2] integrate repair-crew scheduling and routing with relief distribution directly, and Iloglu and Albert [3] co-optimize recovery with emergency response. Repair sequencing is itself well studied as an integrated network design and scheduling problem [4], [5], including debris clearance under incomplete damage information [6], accessibility-oriented crew scheduling [7], and inspection-before-repair resilience objectives [8]. The broader disaster-operations literature [9], [10], [11] covers facility location, inventory, and vehicle routing under uncertainty, and last-mile relief distribution has its own body of work [12] in which the choice of objective—deprivation or unmet demand rather than logistics cost alone—is known to change the delivered allocation [13], [14]. Progressive (staged) recovery, and the time-dependent resilience metrics used to evaluate it [15], [16], [17], appear in several infrastructure-restoration models; heterogeneous fleets are standard in relief routing; and link-capacity constraints are common in traffic assignment. However, these mechanisms are usually adopted as fixed modeling choices rather than as objects of study whose decision relevance is itself diagnosed.

b) Multi-objective evolutionary optimization.: NSGA-II [18] remains the workhorse for multi-objective combinatorial problems, valued for its elitist non-dominated sorting and crowding-distance diversity; the elitist-archive line it belongs to [19] established the dominance-based selection and external archive our framework relies on. Performance assessment relies on indicators such as hypervolume (HV) and inverted generational distance (IGD) [20]. Relief logistics has a long multi-objective tradition—three-objective delivery-system design [21] and equity/efficiency/efficacy routing comparisons [14] predate NSGA-II's dominance in this area—and recent work applies NSGA-II and hybrids (e.g., with adaptive large-neighborhood search) to relief-distribution trade-offs among unmet demand, cost, and fairness, including joint relief distribution and short-term network restoration [22]. These studies concentrate on *how to optimize* a given model, not on *when the model's complex mechanisms change the answer*.

c) Parallel evolutionary and simulation-based optimization.: Parallel genetic algorithms are well surveyed [23], [24]: the dominant paradigms are master–slave (distributed fitness evaluation), island models (distributed populations with migration), and fine-grained diffusion models. For NSGA-II specifically, master–slave schemes have been studied both synchronously and asynchronously, trading synchronization overhead against per-evaluation parallel efficiency [25], [26], and asynchronous master–slave designs have been shown to preserve front quality in other multi-objective metaheuristics [27]. For simulation-based optimization, master–slave parallelization of fitness evaluation is the natural fit because a single evaluation—here, a full multi-period network simulation—dominates runtime while the evolutionary operators themselves are cheap; coupling a metaheuristic to expensive simulated evaluation is itself a studied design pattern [28]. Existing applications rarely combine this with a diagnosis of *which* expensive model components are actually decision-relevant, which is precisely the combination this paper pursues: mechanism-aware analysis decides what fidelity is worth paying for, and parallel evaluation reduces the cost of paying for it.

TABLE I
HETEROGENEOUS VEHICLE FLEET (WEN38 CALIBRATION).

Type	Payload (t)	Fleet size	Min. progress	PCU
1	5	150	0.30	1.0
2	10	110	0.50	1.5
3	15	70	0.70	2.0
4	20	40	0.80	2.5

TABLE II
PROGRESSIVE RECOVERY STAGES.

Progress p	Stage	Capacity / speed ratio
$[0, 0.30)$	blocked	0
$[0.30, 0.60)$	temporary	0.30
$[0.60, 0.80)$	one-lane	0.60
$[0.80, 1)$	basic	0.80
1	full	1.00

III. PROBLEM FORMULATION

We build on the multi-period road repair and relief logistics model of Li and Teo [1], extended with staged recovery, vehicle-specific accessibility, and per-period edge throughput. Evaluation semantics follow our v2 contract: dispatches in each period use *period-start* road conditions, trips longer than one period do not count as current-period arrivals, and no global supply–demand ratio caps per-node allocation.

A. Road network, horizon, and damage

Let $G = (V, E)$ be an undirected road network. The Wenchuan case (WEN38) has $|V| = 38$ nodes, $|E| = 51$ edges, of which $|D| = 16$ edges are damaged; 3 nodes are supply depots and 35 are demand nodes, with total supply $9{,}000$ t against total demand $11{,}288$ t (maximum achievable satisfaction $\approx 79.73\%$). The planning horizon spans $T = 9$ periods of $\eta = 8$ h (72 h total). A set of $K = 3$ repair crews restores damaged edges; each crew works on one edge at a time with a known repair duration.

B. Heterogeneous vehicle fleet

Four vehicle types differ in payload, fleet size, minimum recovery progress required for passage, and passenger-car-unit (PCU) footprint (Table I). A vehicle may traverse an edge only if the edge's recovery progress meets its type-specific threshold, and each vehicle makes at most one trip per period.

C. PR: progressive recovery

A damaged edge's recovery progress $p \in [0, 1]$ maps to discrete stages with capacity/speed ratios (Table II). The binary-recovery counterpart (used in ablations) admits only $p \in \{0, 1\}$.

D. HT: heterogeneous accessibility threshold

Given edge progress p, the passable vehicle set is $\{k : \theta_k \leq p\}$ with thresholds θ_k from Table I. Disabling HT (ablation) unifies all thresholds to 0.30 while keeping payloads, PCU footprints, and trip budgets heterogeneous.

E. EC: edge-period capacity

Each edge-period admits bounded throughput measured in PCU: $\sum \text{trips} \times \text{PCU}_k \leq \text{cap}(e, t)$, where $\text{cap}(e, t)$ scales with the edge's recovery stage. When capacity is exhausted, dispatch reroutes on the residual-capacity graph. Disabling EC removes throughput accounting but keeps passability.

F. Decision encoding and simulation

A candidate solution is a triple $x = (\pi, a, \sigma)$: a repair sequence π over damaged edges, a crew assignment a mapping each crew to its ordered edge list, and a dispatch priority order σ over supplier–demand pairs. Decoding simulates period by period:

1) Advance repair progress from crew schedules; derive each edge's stage and capacity/speed ratio (Table II).
2) Build vehicle-specific passability graphs from *period-start* conditions using thresholds θ_k (HT).
3) Allocate relief by a max-min service-level rule, then allocate remaining servable demand; each allocation respects remaining supply, remaining demand, per-type trip budgets, passability, single-trip time $\leq \eta$ (within-period arrival), and edge-period PCU capacity, with rerouting on the residual-capacity graph when EC binds (EC).

Supply and demand carry over across periods; vehicle spatial positions and return trips are not tracked beyond the exogenous per-period trip budget, a coarse discrete-time approximation consistent with the strategic planning level of the model.

G. Model variants for ablation

To isolate mechanism effects we define three ablated variants of the full model. **No-PR** replaces staged recovery with binary recovery ($p \in \{0, 1\}$), removing all intermediate states. **No-HT** unifies all passability thresholds to 0.30 while keeping payloads, PCU footprints, and trip budgets heterogeneous, so only the threshold structure is removed. **No-EC** drops edge-period throughput accounting but retains passability. Toggling exactly one mechanism at a time yields paired comparisons in which any change in (F_1, F_2, F_3) is attributable to that mechanism alone.

H. Objectives

We minimize three objectives computed by the shared evaluator `evaluate_capacity_solution()`:

$$F_1 = \textstyle\sum_{t=1}^{T} \big(1 - s_t\big), \tag{1}$$

$$F_2 = T_{\text{travel}} + w_r \left(T_{\text{repair}} + T_{\text{transfer}}\right), \tag{2}$$

$$F_3 = -\min_i s_i^{\text{final}}, \tag{3}$$

where s_t is the total demand satisfaction at the end of period t, s_i^{final} the final satisfaction of demand node i, and $w_r = 0.05$ the default repair-time weight. F_1 is therefore measured in unmet-demand proportion–periods, F_2 in weighted minutes, and F_3 maximizes the worst-served node's final satisfaction. Objective comparisons use fixed service resolutions $(10^{-8}, 10^{-6}, 10^{-8})$

with quantized comparison keys so that floating-point noise cannot create spurious Pareto points.

IV. Mechanism-Aware Parallel Optimization Framework

Our framework has two complementary parts: a *mechanism diagnosis* that determines when PR/HT/EC genuinely affect decisions, and a *parallel multi-objective optimizer* that makes full-fidelity evaluation affordable.

A. Mechanism-aware analysis: from existence to effect

A common pitfall is to conflate "the mechanism is in the model" with "the mechanism changes the decision." We separate three evidence layers of increasing strength:

L1 – Exposure (necessary, not sufficient). *Structural/dynamic exposure*: the mechanism *could* matter, e.g. a damaged edge sits in a partially recovered state on some period-start snapshot (PR), a period-start progress admits a *proper nonempty subset* of vehicle types (HT), or an edge's throughput utilization is high (EC).

L2 – Dispatch (stronger). *Dispatch participation*: toggling the mechanism off changes the actual dispatch—routes, vehicle types, tonnages, or trip counts—for a fixed decision.

L3 – Objective (definitional). *Objective effect*: toggling the mechanism changes the quantized objective vector (F_1, F_2, F_3).

Two logical relations, verified over 360 decision–mechanism pairs across network scales, govern how these layers may be used. *Objective* $\Rightarrow$ *dispatch* holds without exception: if the objectives changed, the canonical dispatch encoding necessarily changed, so dispatch comparison is a safe *screening* criterion with no false negatives. The converse fails: on the 100-node network we observed three cases where the dispatch changed yet all three objectives were bit-identical—closing HT made two vehicle types interchangeable in passability while EC was non-binding, so the dispatcher merely swapped vehicle labels between identical shipments. Allocation-change is therefore a *conservative* probe (it over-reports), never the definition; the definition of "the mechanism matters" is always the objective comparison.

Diagnosis proceeds through three probe families. *Fixed-decision toggle probes* evaluate a fixed set of decisions (one SPT-constructed plus random draws) with each mechanism switched on and off, recording exposure, dispatch, and objective indicators per decision. *Resource grids* sweep fleet, supply, and capacity-scale multipliers to locate regimes where each mechanism binds, and *bottleneck relaxation probes* strictly relax one resource at a time (fleet $\times 2$, supply $\times 1.5$, road capacity $\times 2$) to attribute binding resources—a strictly more reliable test than inspecting leftover supply. Finally, *zone calibration* bisects the capacity scale per (scenario, seed) so that measured edge utilization lands in targeted inactive/transition/binding bands, because fixed scales cannot place different seeds inside a narrow transition band. The resulting applicability conditions are summarized in Table III and detailed in Section V.

TABLE III
Applicability conditions of PR / HT / EC, measured over 20 fixed decisions on WEN38 and 1,120 resource-grid cells across S025, S050 and M100.

Condition	Finding
PR binds only if a damaged edge lies on an *actually used* delivery path while in a partial-recovery state at period start.	Exposure window jointly governed by (repair time, crew count, period length); longer repairs *shrink* exposure under fixed crews.
HT binds only if some edge's period-start progress admits a proper nonempty subset of vehicle types on a used path with remaining trips. A bridge/corridor is *sufficient, not necessary*.	Node-role layout alone raised participation 2/20 $\rightarrow$ 6/20 with zero damaged bridges.
EC binds only at high edge utilization: < 0.85 never (0/776 cells across three scales); 0.85–0.97 transition ($\approx$46%); $\geq$0.97 binding ($\approx$99%). Sharper: rerouting occurred $\Rightarrow$ binding (296/296).	Baseline calibration (utilization 0.010–0.034) is far from binding; non-effect there is a calibration property, not model failure.

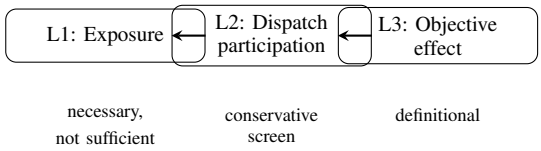


Fig. 1. Three-layer mechanism diagnosis. Objective change always implies dispatch change; the reverse can break (e.g., vehicle-interchange ties), so L2 screens but never defines.

In non-binding zones, adding the corresponding mechanism cannot significantly change solutions—this is an *applicability boundary*, not a failure of the model.

B. Parallel multi-objective optimization

Even where mechanisms matter, each fitness evaluation runs the full T-period simulation of Section III, whose cost grows with network size. Since candidate evaluations are mutually independent given the population, we parallelize at the population level with a master–worker design (Fig. 2).

a) Parallel: fitness evaluation.: The master generates an offspring batch; each candidate is dispatched to a worker that runs `evaluate_capacity_solution()` end to end and returns (F_1, F_2, F_3). Workers are stateless with respect to the search: they hold no population, no archive, and no RNG state beyond a fixed evaluation seed.

b) Centralized: everything else.: Selection (non-dominated sorting, crowding distance), crossover, mutation, random-number generation, external archiving, and budget accounting all remain on the master. The evaluation *budget* counts evaluations, not generations or wall-clock time, so search effort is comparable across worker counts.

c) Reproducibility guarantee.: Candidates are assigned to workers deterministically and results are reassembled by candidate index before ranking; all stochasticity flows from

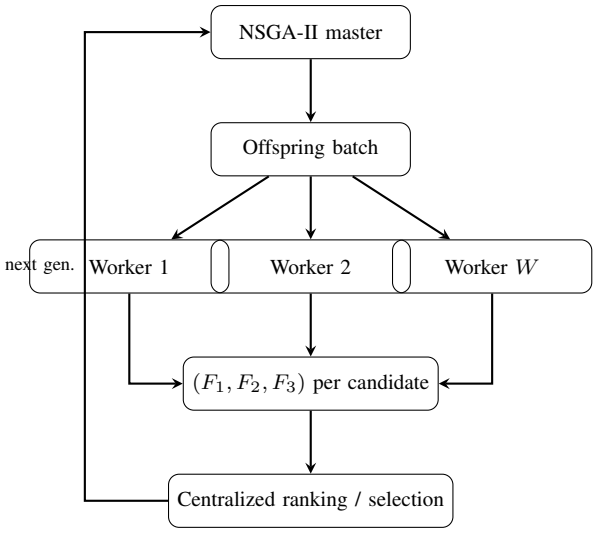


Fig. 2. Master–worker parallel framework. Only fitness evaluation is distributed; selection, crossover, mutation, RNG, archiving, and budget accounting stay centralized, preserving exact reproducibility.

Protocol: master–worker parallel NSGA-II

```
master(seed):
 rng = init_central_rng(seed)
 P = init_population(rng)
 evaluate_parallel(P)
 archive = nondominated(P)
 while budget > 0:
  Q = select_cross_mut(P, rng)
  F = evaluate_parallel(Q)
  P = environmental_sel(P, Q)
  archive = update(archive, P)
  budget -= |Q|

evaluate_parallel(C):
 for i, x in enumerate(C):
  send (i, x) to worker i mod W
 collect; reassemble by index i
 return [(F1,F2,F3)] per cand.

worker(i, x):
 return evaluate_capacity_
        solution(x)
 # full T-period simulation;
 # no search state, no RNG
```

Fig. 3. Parallel evaluation protocol. Workers are stateless; identical seeds yield identical Pareto fronts for any worker count W.

the master's centralized RNG streams. Consequently, identical seeds produce *bit-identical* evaluated populations—and hence identical Pareto fronts—for any number of workers W. Parallelism changes *when* results arrive, never *what* the optimizer decides. We verify this protocol invariance empirically in Section V (RQ3): HV/IGD across $W \in \{1, 2, 4, 8\}$ must match exactly, while wall-clock time, speedup $S(W) = T_1/T_W$, and efficiency $E(W) = S(W)/W$ quantify the savings.

This separation is deliberate. Island models exchange migrants and thereby change the search trajectory; our design keeps the trajectory fixed and parallelizes only the expensive, independent simulations. The mechanism-aware analysis of Section IV-A complements it: it tells us in which zones the full-fidelity evaluation is worth its cost, while parallelism lowers that cost everywhere.

d) Cost structure.: Each evaluation simulates T periods; per period it rebuilds vehicle-specific passability graphs (shortest paths over $|E|$ edges) and runs capacity-constrained, vehicle-aware dispatch over supplier–demand pairs. The dominant terms therefore grow with network size and fleet size, while master-side non-dominated sorting costs $O(MN^2)$ for population N and $M = 3$ objectives. The parallelizable fraction is therefore governed by how much work one dispatch carries relative to the master's serial share, while the number of independent evaluations available per generation—at most the population size—caps the parallel width itself. RQ3 and RQ4 measure both effects and confirm that the second, not the first, is what limits scaling here.

V. Experimental Evaluation

We structure evaluation around four research questions:

RQ1. Do PR, HT, and EC materially affect recovery–logistics decisions?

RQ2. Under what conditions do these mechanisms become active or binding?

RQ3. Can parallel evaluation reduce runtime without changing the optimization protocol?

RQ4. What governs the parallel efficiency of population evaluation across instance scales and worker counts?

a) Instances and protocol.: We use the Wenchuan WEN38 network and the multi-scale synthetic suite (S025, S050, M100): synthetic instances default to a 0.9 supply–demand ratio and capacity scale 0.05, with fleet sized at $0.14\times$ total demand per period of nominal capacity. The optimizer is NSGA-II; model-ablation studies use NSGA-II to isolate model effects. Quality is measured by normalized HV and Euclidean IGD against pooled reference fronts, with paired main effects, 95% BCa intervals, and Holm correction; solver repeats are averaged within an instance before instances are equally weighted. Ablations toggle one mechanism at a time: No-PR uses binary recovery, No-HT unifies thresholds to 0.30, and No-EC drops edge-period throughput. Every saved decision is additionally replayed under the common Full execution to separate *planning-front quality*, *selected-solution execution*, and *planning-vs-execution prediction bias*.

A. RQ1: Do the mechanisms materially affect decisions?

Paired ablations show the answer is mechanism- and zone-dependent rather than a blanket yes. We run 56 paired NSGA-II searches (500 evaluations, population 32): WEN38 with two solver repeats across the four model variants, and the S025 corridor across two instances, three capacity zones, and two solver repeats. On WEN38 this yields PR comparisons favoring the Full model, an HT front-quality reversal across the two repeats, and zero EC effect at the baseline calibration—consistent with EC's diagnosed non-binding zone there (Section IV-A). With two repeats per cell these are paired contrasts, not effect-size estimates; the zone-resolved replay below is what carries the argument.

TABLE IV
COST OF IGNORING A MECHANISM: ABLATED-MODEL DECISIONS REPLAYED UNDER COMMON FULL EXECUTION (S025 CORRIDOR, 813 REPLAYED DECISIONS). "CHANGED" COUNTS DECISIONS WHOSE OBJECTIVES MOVED, OUT OF THOSE REPLAYED.

Zone	Plan model	Changed	Mean ΔF_1	Max $\lvert\Delta F_2\rvert$ (min)
inactive	No-PR	33/37	-0.798	2,663
	No-HT	55/59	$+0.275$	934
	No-EC	**0/93**	0.000	0
transition	No-PR	32/38	-0.166	1,039
	No-HT	33/53	$+0.129$	295
	No-EC	49/93	$+0.457$	3,659
binding	No-PR	53/58	-0.309	745
	No-HT	55/67	$+0.034$	302
	No-EC	**93/93**	$+0.691$	3,783

TABLE V
EC BINDING VS. MAXIMUM EDGE UTILIZATION (1,120 CELLS; S025/S050/M100).

Max utilization	S025	S050	M100	Combined
< 0.85	0/517	0/151	0/108	**0/776**
0.85–0.97	17/35	5/10	0/3	22/48 (46%)
$\geq$0.97	88/88	79/79	125/129	292/296 (99%)
Rerouting criterion (all scales):				
reroutes > 0				296/296 (100%)
reroutes $= 0$				18/824 (2%)

The cost of ignoring a mechanism is quantified by replaying each ablated model's planned decisions under the common Full execution (Table IV, S025 corridor, 813 replayed decisions). Ignoring EC where it binds underestimates unmet demand by up to 2.73 proportion–periods and misestimates time cost by up to 3,783 min, while in the inactive zone No-EC plans are mathematically identical to Full (0/93 changed)—exactly the non-binding definition. Ignoring PR changes most decisions in all zones (33/37, 32/38, 53/58), concentrated in F_2 (mean $+$333–1,570 min), because one critical road's repair timing dominates execution cost. Ignoring HT incurs smaller losses (mean $+$10–115 min in F_2). Worst-node fairness (F_3) also moves in most affected decisions, confirming that mechanism effects are not confined to aggregate objectives.

B. RQ2: When do the mechanisms become active or binding?

a) EC: utilization bands with a sharper rerouting criterion.: Across 1,120 grid cells (three scales, two topology families), EC never alters objectives below 0.85 maximum edge utilization (0/776), binds in about half the 0.85–0.97 transition cells (22/48), and binds in $\approx$99% of cells at $\geq$0.97 (292/296); see Table V. A sharper sufficient criterion is *capacity-induced rerouting*: cells with reroutes bind in 296/296 cases. The 0.97 edge is not a hard threshold (four M100 cells above it do not bind), so zone selection must calibrate by measured utilization rather than fixed scales.

b) PR: exposure window governed by crew resources.: PR exposure requires a damaged edge on an actually used path in a partial state at period start. Counter-intuitively, under a fixed crew count and 72 h horizon, lengthening repair times *reduces* exposure—more roads never leave the blocked stage—so the window is jointly controlled by (repair time, crew count, period length), not by duration alone.

c) HT: corridor sufficient, not necessary.: HT needs a used edge whose period-start progress admits a proper nonempty subset of vehicle types. A damaged bridge is one sufficient topology, but node-role layout alone raised HT participation from 2/20 to 6/20 with zero damaged bridges. On the real WEN38 topology with semi-realistic overlays, a four-level funnel confirms a complete chain: 15/16 damaged edges are structurally relevant, 5–21 edge–periods per decision show dynamic exposure, and 19/20 decisions change both dispatch and objectives when HT is toggled.

d) Resource bottlenecks.: Relaxation probes (fleet $\times$2, supply $\times$1.5, road capacity $\times$2, each strictly a relaxation) show the fleet binds in both scenario families, with supply co-binding under random damage—there is no single bottleneck, and "all supply delivered" alone cannot identify one.

C. RQ3: Parallel runtime reduction without protocol change

We verify two claims: (i) *protocol invariance*—identical seeds yield identical fronts for every worker count—and (ii) *wall-clock savings*—speedup and efficiency of population evaluation. Table VI reports full-budget NSGA-II runs on the real WEN38 instance: population 32, a 1,600-evaluation budget, eight solver seeds, and $W \in \{0, 1, 2, 4, 8\}$. The $W{=}0$ row is the in-process serial path, which we first verified to be bit-identical to the repository's reference NSGA-II implementation at a smaller budget, so it serves as the protocol baseline rather than merely as a timing reference.

Invariance is exact. Every run consumed its full evaluation budget and produced a Pareto front *bit-identical* to the serial run of the same seed—fronts of 30 to 72 points across the eight seeds—with a maximum absolute objective deviation of exactly zero over all 40 runs and matching objective-cache hit counts. Because the fronts coincide, pooled HV and IGD are identical across W by construction and are therefore not tabulated per worker count. Parallelism changes *when* results arrive, never *what* the optimizer decides, as the reproducibility guarantee of Fig. 3 requires.

Wall-clock savings are near-linear through $W{=}4$ and then saturate: speedup against the serial baseline is $1.78\times$, $3.00\times$ and $3.18\times$ at $W = 2, 4, 8$ (efficiency 0.90, 0.76, 0.40). The $W{=}1$ row is marginally *slower* than serial ($0.99\times$): a single worker pays process and IPC overhead without recovering any parallelism. We report both baselines throughout, which makes the $W{=}1$-referenced figures conservative and keeps RQ4 comparable, where no $W{=}0$ runs exist.

The saturation at $W{=}8$ is a property of the parallel *grain*, not of the machine—eight workers on thirteen cores are not oversubscribed. Parallel width is bounded by the generation batch: with population 32 a generation offers at most 32 independent evaluations, fewer once objective-cache hits remove unmutated clones, and each evaluation is only $\approx$46 ms of work

TABLE VI
FULL-BUDGET PARALLEL NSGA-II, WEN38 (POP. 32, 1,600 EVALUATIONS); MEAN ± S.D. OVER EIGHT SOLVER SEEDS (11,22,33,44,55,66,77,88). $S_0 = T_0/T_W$ AND $S_1 = T_1/T_W$ ARE SPEEDUPS AGAINST THE IN-PROCESS SERIAL PATH AND THE SINGLE-WORKER PATH; $E(W) = S_1/W$. ALL 40 FRONTS ARE BIT-IDENTICAL TO THEIR SERIAL RUN.

W	Time (s)	S_0	S_1	$E(W)$
0 (serial)	72.1±2.3	–	–	–
1	73.1±2.1	0.99×	1.00×	1.00
2	40.6±2.1	1.78×	1.80×	0.90
4	24.1±1.4	3.00×	3.04×	0.76
8	22.7±1.7	3.18×	3.23×	0.40

at WEN38 scale, so per-generation synchronization and dispatch overhead become comparable to the useful work, leaving the master's serial share (non-dominated sorting, selection, operator application, RNG) as the residual floor. The design consequence is that efficiency should *rise* with per-evaluation cost, which RQ4 tests directly.

D. RQ4: What governs parallel efficiency

RQ3 left an open question: if parallel width is bounded by the generation batch, what determines how much of that bound is actually reachable? RQ4 answers it across the instance ladder S025 (25 nodes), WEN38 (38), S050 (50) and M100 (100), at per-case budgets of 1,600, 1,600, 800 and 400 evaluations, all with population 32 and eight solver seeds (Table VII). Budgets shrink with instance size to keep the study tractable, so the larger rows measure parallel cost rather than converged search quality.

Invariance holds exactly again: within every (instance, seed) block all four worker counts produce the same front—0 deviations across the 128 runs at $W \in \{1, 2, 4, 8\}$, and across all 136 runs. Front sizes vary with the seed but never with W (S025: 2–6; WEN38: 30–72; S050: 4–17; M100: 9–24), as do the objective-cache hit counts; the pooled HV and IGD are likewise invariant in W and are reported once per instance rather than per worker count. On M100, for example, seed 11 gives HV $= 0.7049$ and IGD $= 0.2204$ at every $W \in \{1, 2, 4, 8\}$.

Efficiency tracks evaluation *granularity*, not network size. Measured at W=1, a single dispatch carries 46 ms on WEN38 and 49 ms on the *smaller* S025, but 197 ms on S050 and 618 ms on M100 (Table VII). The corresponding W=8 efficiencies are 0.40, 0.43, 0.56 and 0.61, and the rank order of the four instances by efficiency matches their rank order by granularity at every W—the two fine-grained instances lowest, the two coarse-grained highest, with no inversion at any worker count. Two instances of near-identical granularity therefore reach near-identical efficiency despite a $1.5\times$ difference in size, while the $13.5\times$ larger granularity on M100 buys a 52% higher W=8 efficiency. Efficiency still declines with W everywhere, because the independent evaluations available per generation—at most the population size, and fewer once cache hits remove unmutated clones—bound the parallelizable fraction no matter how large the instance grows.

TABLE VII
SCALE LADDER: POPULATION 32, EIGHT SOLVER SEEDS, $W \in \{1, 2, 4, 8\}$; MEANS ± S.D. OVER SEEDS. WITHIN EVERY (INSTANCE, SEED) BLOCK ALL WORKER COUNTS YIELD IDENTICAL FRONTS.

Instance	W=1	W=2	W=4	W=8
Wall-clock time (s)				
S025 (25)	78.0±0.7	42.6±1.0	24.3±1.4	22.7±0.8
WEN38 (38)	73.1±2.1	40.6±2.1	24.1±1.4	22.7±1.7
S050 (50)	157.4±2.1	82.0±1.7	43.5±0.7	35.0±0.6
M100 (100)	247.1±1.6	126.4±0.8	66.6±0.9	50.4±0.7
Speedup $S(W) = T_1/T_W$				
S025 (25)	1.00×	1.83×	3.20×	3.44×
WEN38 (38)	1.00×	1.80×	3.04×	3.23×
S050 (50)	1.00×	1.92×	3.62×	4.49×
M100 (100)	1.00×	1.96×	3.71×	4.90×

This sharpens the design guidance and connects the two halves of the paper. Additional workers pay off in proportion to the cost of a single evaluation, so the lever for an expensive instance is raising the search budget—more evaluations, hence more independent work per generation—rather than adding workers to an instance whose evaluations are already cheap. The mechanism-aware diagnosis of Section IV-A pulls in the opposite direction: in a diagnosed non-binding zone, a reduced-fidelity evaluation can replace the full one at zero decision cost, which lowers granularity and therefore *reduces* the marginal value of each additional worker. Fidelity reduction and parallel evaluation are thus substitutes, not complements, and which one is appropriate in a given regime is exactly what the applicability conditions of Section IV-A answer.

a) Validity and limits.: The RQ3/RQ4 results are full-budget NSGA-II measurements over 136 runs with frozen seeds, but they are wall-clock measurements on one shared 13-core machine; absolute times are machine-specific, and the W=1-referenced speedups should be read as protocol-level ratios rather than as portable constants. RQ4 uses eight solver seeds per instance and budgets that shrink with instance size, so its larger rows characterize parallel cost, not converged search quality. RQ1/RQ2 are diagnostic rather than comparative: fixed-decision toggles, resource grids and relaxation probes establish *where* each mechanism can bind, but their sample sizes—two solver repeats per ablation cell, 20 fixed decisions on WEN38—support applicability boundaries, not effect sizes or significance claims, which would need multi-instance, multi-seed protocols. Stress multipliers are diagnostic, not calibrated field parameters.

VI. CONCLUSION

We studied not only how to optimize a realistic post-disaster recovery–logistics system, but also when its complex physical mechanisms matter and how their expensive evaluations can be efficiently parallelized. Four findings summarize the paper: (1) complex mechanisms are not always binding—PR, HT, and EC each have identifiable non-binding, transition, and

binding zones; (2) mechanism applicability conditions explain *when* the full model is worth its cost, turning fidelity selection from guesswork into verifiable conditions; (3) master–worker parallel evaluation cuts the wall-clock cost of expensive multi-period fitness evaluation without changing the optimization protocol—over 136 full-budget NSGA-II runs spanning 25 to 100 nodes, every worker count returned a Pareto front bit-identical to its reference run (in-process serial on WEN38, one worker on synthetic instances), with speedups up to $4.9\times$; and (4) the resulting efficiency is governed by evaluation granularity rather than network size, so fidelity reduction and parallel evaluation substitute for one another and should be deployed in opposite regimes. Future work includes distributed execution beyond a single node, larger-scale networks, tighter integration of the applicability conditions with the choice between fidelity reduction and additional workers, and dynamic rolling-horizon optimization under revealed repair progress.

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